PV102 Day 6: Inverter Charge Controller and Protection Boundaries

Students map the balance-of-system decisions that make PV energy usable and controlled: charge controllers, inverters, disconnects, overcurrent protection, grounding/bonding boundaries, and labeled handoff points.

PV102 | CTS Day 22 | Instructor Run Sheet
Power electronics set system boundaries. | Tuesday, August 11, 2026

Core rule

Power electronics set system boundaries.

Learning targets

- Explain the roles of charge controllers and inverters in battery-based PV systems.
- Match inverter selection to continuous load, surge load, voltage, and output requirement.
- Identify where disconnects and overcurrent protection belong conceptually.
- Create a block diagram that separates source, storage, conversion, and load boundaries.

Readiness and materials

Before students arrive	Materials
Open slide deck and test display. Print student packets and one instructor key. Prepare three team stations. Review safety boundary before any equipment is handled.	Component cards: PV, controller, battery, inverter, OCPD, disconnect, load String/yarn or arrows Calculators Sample labels Projector and slide deck

Five-hour run of show

Time	Block	Instructor move
10:00-10:15	Opening and continuity	Connect Day 5 to today's rule. Name what students already know and what new judgment is being added.
10:15-10:40	Morning quiz and review	Students work independently for 12-15 minutes. Review by asking for evidence, not just letters.
10:40-11:25	Concept lesson	Use slides 1-14. Pause after each technical rule and ask what decision it changes.
11:25-12:15	Guided practice	Model one example on the board. Students complete a quick pair-check before lunch.
12:15-12:45	Lunch/reset	Reset lab stations. Keep energized work out of scope unless explicitly authorized by instructor and equipment procedure.
12:45-1:00	Lab brief	Assign teams and roles. Review

		acceptance criteria and stop-work triggers.
1:00-2:15	Balance-of-System Block Diagram	Teams build a labeled block diagram and defend every protection/isolation point.
2:15-2:40	Handoffs and peer verification	Each team gives a concise technical handoff. Peers check evidence, boundary, and unknowns.
2:40-3:00	Exit ticket and reset	Collect packets. Preview next step: PV102 Day 7 turns the design into an install plan, tool/material list, JHA, and daily work sequence.

Instructor talk track

Opening script: "PV102 is not about throwing parts at a problem until reality files a complaint. Today we use inverter charge controller and protection boundaries to make the next design decision safer and more defensible. The rule is simple: Power electronics set system boundaries."

Teaching notes

Move	Look-for
Ask	What decision does this evidence change?
Listen for	Students separating facts, assumptions, and unknowns.
Correct gently	Component-first answers. Redirect to job, evidence, boundary, then component.
Require	A handoff that another person could act on without reading minds. Humanity keeps trying telepathy. It remains a poor project-management system.

Lab instructor procedure

- Choose a scenario and place component cards in energy-path order.
- Draw source, storage, conversion, and load boundaries.
- Add conceptual OCPD/disconnect labels where energy could feed a fault or where service isolation is needed.
- Mark manufacturer-instruction-required decisions.
- Run a peer safety check: can someone tell what is energized, isolated, and unknown?
- Handoff the diagram and unresolved decisions.

Answer snapshot and contingency plan

If equipment or printing is unavailable, run the lab with paper cards, chart paper, and oral handoffs. The learning target is decision quality, not hardware theater.

Safety and scope boundaries

No live electrical work is authorized by this lesson unless the instructor independently sets up a verified training system, follows manufacturer instructions, and establishes the required isolation, PPE, and supervision. Treat PV source circuits and batteries as energy sources, not props.

Day handoff

PV102 Day 7 turns the design into an install plan, tool/material list, JHA, and daily work sequence.

Instructional sources

- U.S. Department of Energy, Solar Energy: https://www.energy.gov/topics/solar-energy
- U.S. Department of Energy, Solar Photovoltaic Technology Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-technology-basics
- U.S. Department of Energy, Solar Photovoltaic System Design Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-system-design-basics
- U.S. Department of Energy, Solar Integration: Solar Energy and Storage Basics: https://www.energy.gov/eere/solar/solar-integration-solar-energy-and-storage-basics
- U.S. Department of Energy, Solar Integration: Inverters and Grid Services Basics: https://www.energy.gov/eere/solar/solar-integration-inverters-and-grid-services-basics
- NREL PVWatts Calculator: https://pvwatts.nrel.gov/
- OSHA, Green Job Hazards - Solar Energy: Electrical: https://www.osha.gov/green-jobs/solar/electrical
- OSHA, Green Job Hazards - Solar Energy: Lockout/Tagout: https://www.osha.gov/green-jobs/solar/lockout-tagout
- OSHA, Green Job Hazards - Solar Energy: Personal Protective Equipment: https://www.osha.gov/green-jobs/solar/personal-protective-equipment
- Sean White, Solar Photovoltaic Basics, 2nd ed., Routledge, 2018, ISBN 9781138102866.